

BTECH
(SEM V) THEORY EXAMINATION 2022-23
GENETIC ENGINEERING

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. 2 x 10 = 20

- a. Define Cosmids, Phagemids, Methylases and Phosphatase.
- b. What is the process of DNA isolation?
- c. Differentiate between jumping and cloning libraries.
- d. What is blue-white screening?
- e. Which are the key features required for a DNA polymerase for polymerase chain reaction?
- f. What is the principle of PCR primer designing?
- g. How DNA is used as therapeutics?
- h. Draw a flow chart of cloning of sheep – Dolly.
- i. Differentiate between intercellular and intracellular cell signaling.
- j. Enlist the characteristics of the receptor- tyrosine kinases.

SECTION B

2. Attempt any three of the following: 10 x 3 = 30

- a. Human CFTR gene is 250KB. Choose an ideal vector to clone this gene and alsodescribe the construction and screening principles of that vector.
- b. Write short note on-
 - i. Chromosome walking
 - ii. BACs
- c. Explain in detail the principal, procedure and application of PCR. Add a note on real time PCR.
- d. Write various methods involved in production of insecticide resistant transgenic plants.
- e. What are different types of cell-signaling? Enlist characteristics of each type.

SECTION C

3. Attempt any one part of the following: 10 x 1 = 10

- a. What are restriction endonucleases? Explain in detail the various types.
- b. Write short note on-
 - i. Phage as vector
 - ii. Synthetic promoter used in expression vector
 - iii. Alpha complementation

4. Attempt any *one* part of the following: 10 x 1 = 10
- a. Describe the construction of a genomic library. What are its applications?
 - b. Describe the production of monoclonal antibodies by hybridoma technology.
5. Attempt any *one* part of the following: 10 x 1 = 10
- a. Write short note on-
 - i. Nested PCR
 - ii. Taqman Assay
 - iii. Maxam Gilbert method of DNA sequencing
 - b. Evaluate the efficiencies and deficiencies of RFLP, RAPD and SNP markers.
6. Attempt any *one* part of the following: 10 x 1 = 10
- a. Explain the role of genetic engineering in producing engineered insulin.
 - b. Explain gene targeting and gene silencing.
7. Attempt any *one* part of the following: 10 x 1 = 10
- a. Write note on-
 - i. Intracellular cascades
 - ii. GTPase molecular switch
 - b. Write note on-
 - i. Interconnecting signaling pathway
 - ii. Nitric oxide signaling